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Intelligent Task Allocation and Path Planning for Warehouse Robots in Smart Logistics Systems

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Abstract: Intelligent warehousing has become a key component of Industry 4.0-driven logistics systems, where the coordination of autonomous robots directly affects operational efficiency and system responsiveness. This study addresses the joint optimization of task allocation and path planning for warehouse robots in e-commerce fulfillment environments. A grid-based model is first established to represent the warehouse space, and the scheduling objective is formulated to minimize total travel distance while maintaining balanced workload distribution. An improved genetic algorithm is developed for task allocation, incorporating a multi-layer encoding scheme to represent complex task relationships, along with a simulated annealing mechanism to improve solution quality and prevent premature convergence. For path planning, an enhanced A* algorithm is proposed by introducing a turning cost term into the evaluation function, which effectively reduces unnecessary directional changes and improves path smoothness. Simulation results show that the proposed method significantly outperforms conventional approaches, achieving faster convergence and notable reductions in both travel distance and turning frequency. Specifically, the convergence speed is improved by over 70%, while the total travel distance and the number of turning maneuvers are reduced by approximately 48% and 78%, respectively. The proposed framework enables coordinated decision-making for multi-robot systems and provides a scalable and practically applicable solution for intelligent scheduling in smart logistics environments.

Keywords: Intelligent warehousing; Industrial intelligence; Robot scheduling; Intelligent decision system; Smart logistics

1 Introduction

The change from traditional storage to automation and intelligence has led to the emergence of intelligent warehousing, which is driven by automated equipment integrated with information technologies. Developed countries began research on intelligent warehousing at an early stage. Since the 1950s, Europe and Japan have achieved remarkable progress alongside the application of computer control technologies in automated high-bay warehouses in the United States. The development of intelligent warehousing in China started relatively late. Nevertheless, rapid advancement has been made in recent years driven by the booming e-commerce industry, and relevant technologies have gradually matured [1]. Warehouse robots serve as the core operating units of intelligent warehousing, and their related technologies have undergone substantial evolution. Early intelligent warehousing mainly adopted automated material handling devices, whereas modern intelligent warehousing systems enable intelligent agents to realize fully autonomous navigation and operation. After acquiring Kiva Systems, Amazon launched the "goods-to-person" picking mode in 2012, which greatly improved picking accuracy and operational efficiency, marking an important milestone in warehouse automation [2]. Subsequently, companies such as Swisslog [3] and Fetch Robotics [4] developed robotic systems with distinctive characteristics. Meanwhile, domestic enterprises have actively followed this trend, promoting the application and innovation of warehouse robots in China.

In terms of task allocation for warehouse robots, existing research has mainly focused on centralized and distributed methods. Centralized task allocation relies on a central controller to perform global optimization through common intelligent optimization algorithms, such as genetic algorithms, particle swarm optimization, and simulated annealing. For example, Li et al. [5] optimized task allocation using an improved particle swarm optimization algorithm. Deng et al. [6] enhanced allocation efficiency by combining rolling scheduling with genetic algorithms.

In contrast, distributed task allocation is usually based on market or game theory mechanisms, and tasks are assigned through autonomous negotiation among robots. Representative methods include the auction algorithm proposed by Gerkey and Mataric [7] and the resource auction-based allocation method developed by Zhao et al. [8]. Both methods show strong adaptability in dynamic environments.

Path planning is crucial to ensure efficient and safe task execution for warehouse robots. A variety of planning methods have been proposed, including graph search, sampling-based methods, bio-inspired intelligent algorithms, and reinforcement learning. For instance, Shang et al. [9] improved obstacle avoidance performance by optimizing the heuristic function of the A* algorithm. Kang et al. [10] improved the Rapidly-exploring Random Tree Connect algorithm based on the triangle inequality, thus enhancing planning efficiency. Wei et al. [11] reduced the number of turns in the planned path through an improved ant colony optimization algorithm. In addition, Zhang et al. [12] adopted deep reinforcement learning to improve the completeness and convergence speed of path search. These studies have provided diverse and effective solutions for robot path planning in complex warehouse environments.

However, several challenges still exist, especially in task balancing, real-time dynamic obstacle avoidance, and multi-robot system optimization. To address these deficiencies, this study conducts joint optimization of task allocation and path planning for e-commerce fulfillment centers. An improved genetic algorithm is adopted for task allocation, in which the introduction of simulated annealing alleviates premature convergence and optimizes the task allocation scheme. For path planning, an enhanced A* algorithm integrating Manhattan distance with a turning cost term is proposed to improve path smoothness. The effectiveness of the proposed method is verified through simulation experiments.

2 Modeling of Intelligent Warehousing System and Problem Description

2.1 System Architecture

An e-commerce intelligent warehousing system is constructed based on the "goods-to-person" picking mode (Figure 1). The system is mainly composed of a central control system, automated guided vehicles, movable storage racks, and manual picking stations. The storage racks are arranged in a grid layout, and material handling tasks are completed by automated guided vehicles under centralized scheduling. Fully automated warehousing logistics equipment such as mobile warehouse robots replace the manual search work previously performed by workers to locate designated storage racks. The central control system issues scheduling instructions, and then warehouse robots are dispatched to designated rack positions to transport the target racks to the picking area. Operators at the picking stations complete order fulfillment tasks by directly picking the required items from the transported racks.

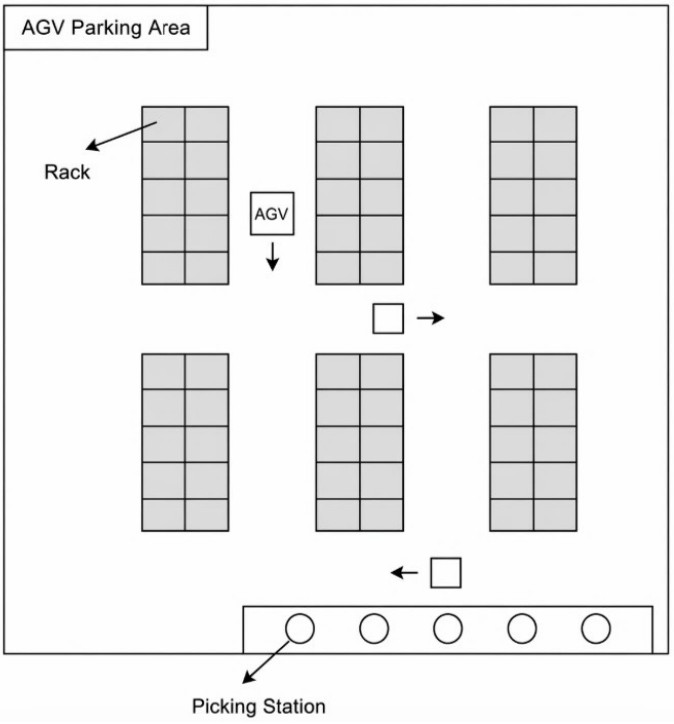


Figure 1. "Goods-to-person" picking diagram

2.2 Grid-Based Environment Modeling

The grid-based method decomposes the workspace into uniform and regularly structured grid cells and models the robot operating environment. In this approach, real-world spatial information is divided into local units (square cells of equal size). Environmental information is encoded within each grid cell using a two-dimensional array or binary matrix. Based on obstacle distribution, grid cells are categorized and assigned corresponding values. For example, 1 denotes cells containing obstacles, and 0 denotes traversable cells [13]. In this study, a grid-based map is employed to model the intelligent warehousing environment. The warehouse layout is a 30 × 30 grid map, in which obstacles (e.g., storage racks and picking stations) are encoded as 1, and navigable aisles are encoded as 0. To ensure feasibility and collision avoidance in path planning, the side length of each grid cell is consistent with the size of the robot (1 m). The intelligent warehousing grid is illustrated in Figure 2.

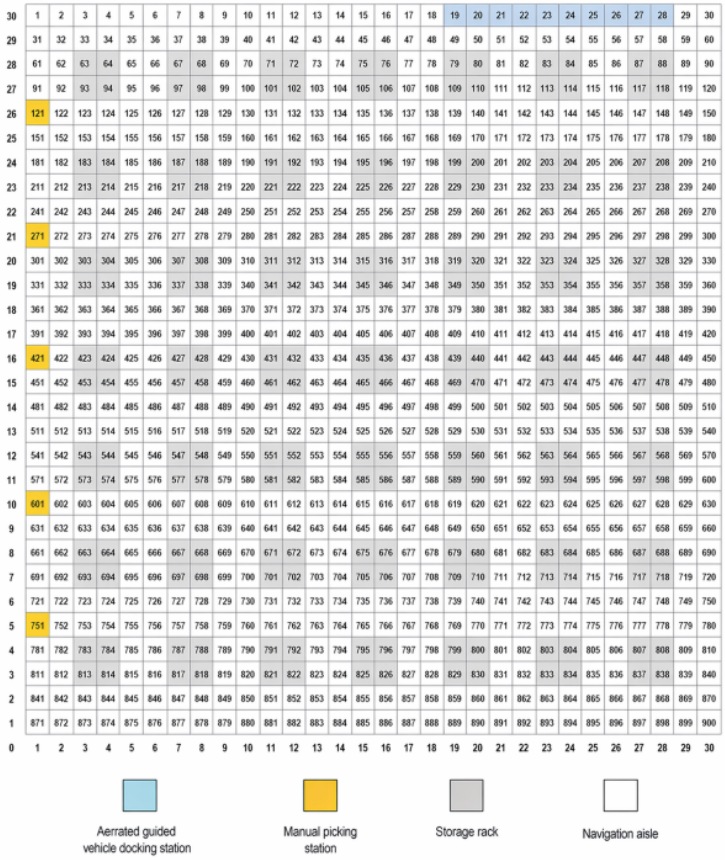


Figure 2. Grid-based intelligent warehousing

2.3 Problem Description

2.3.1 Task allocation problem

In intelligent warehousing, multiple robots collaboratively execute storage rack transportation tasks. Task allocation should minimize the total travel distance and maintain balanced workload distribution among robots. This is the central challenge. The total travel distance is the sum of the round-trip distance between storage racks and picking stations, as well as the inter-task travel distance between successive assignments. The objective is to assign n tasks to m robots to minimize the total travel distance. An improved genetic algorithm is employed. Multi-layer matrix encoding, simulated annealing-based perturbation, and elite preservation are used for optimizing task allocation, effectively preventing premature convergence.

2.3.2 Path planning problem

Each robot should have a global path from its initial position to the designated target location. The planned path should have minimal path length and reduced turning maneuvers. The warehousing environment is modeled as a grid-based map, in which robots only move in four-connected directions (i.e., up, down, left, and right). In the improved A* algorithm, the Manhattan distance and a turning cost term are incorporated to enhance path smoothness.

The effectiveness of the turning cost parameter is validated through simulation experiments. The results are feasible paths with reduced turning frequency and lower energy consumption.

3 Task Allocation Optimization Model and Algorithm

Warehouse robots are the core operational units of intelligent warehousing. They pick and transport items with full automation. The system efficiency depends on optimized task allocation, reduced energy consumption, and flexible scheduling. In this section, a task allocation model is constructed based on practical operational requirements. The improved genetic algorithm optimizes task allocation to minimize total travel distance and enhance picking efficiency.

3.1 Mathematical Model Formulation

In the considered system, a total of m warehouse robots are defined as the set $R = \{r_1, r_2, \ldots, r_m\}$, and a total of n transportation tasks are represented as the set $B = \{b_1, b_2, \ldots, b_n\}$. In addition, k manual picking stations are defined as the set $S = \{s_1, s_2, \ldots, s_k\}$. During the execution of a single task, the travel distance of a warehouse robot is composed of three components: (i) the distance from the current storage rack to the manual picking station, (ii) the distance from the manual picking station back to the original storage rack, and (iii) the distance from the original storage rack to the next target storage rack. Accordingly, the total travel distance for each robot performing a task is defined as follows:

$$d_{rb}^{s} = d_{rb} + 2d_{bs} \tag{1}$$

where, d_{rb}^{s} denotes the total travel distance incurred by robot r when executing transportation task b with the associated manual picking station s; d_{rb} represents the distance from robot r to the location of task b; and d_{bs} denotes the distance between task b and the corresponding manual picking station s.

The total travel distance of all warehouse robots is defined as follows:

$$D = \sum_{r=1}^{m} \sum_{b=1}^{n} \sum_{s=1}^{k} x_{rb}^{s} \cdot y_{rb} \cdot d_{rb}^{s} \tag{2}$$

where, x_{rb}^{s} and y_{rb} are decision variables. Specifically, $x_{rb}^{s} = 1$ indicates that robot r transports storage rack b to manual picking station s; otherwise, $x_{rb}^{s} = 0$. Similarly, $y_{rb} = 1$ indicates that task b is assigned to robot r; otherwise, $y_{rb} = 0$.

With the objective of minimizing the total travel distance of warehouse robots, the task allocation optimization model is formulated as:

$$\min D = \sum_{r=1}^{m} \sum_{b=1}^{n} \sum_{s=1}^{k} x_{rb}^{s} \cdot y_{rb} \cdot d_{rb}^{s} \tag{3}$$

In addition, to satisfy the model assumptions, the following constraints are imposed:

$$\sum_{s=1}^{k} x_{rb}^{s} = 1 \tag{4}$$

$$\sum_{r=1}^{m} y_{rb} = 1 \tag{5}$$

$$\sum_{r=1}^{m} \sum_{b=1}^{n} y_{rb} = m \tag{6}$$

Eq. (4) ensures that each task is assigned to exactly one manual picking station. Eq. (5) guarantees that each task is executed by exactly one warehouse robot. Eq. (6) enforces that all tasks are assigned and executed.

3.2 Improved Genetic Algorithm Design

3.2.1 Encoding and initialization

A three-layer matrix encoding scheme (Table 1) is adopted, in which the three layers correspond to the manual picking stations, warehouse robots, and task sequences, respectively.

The initial population is generated using a randomized initialization strategy. Once the numbers of tasks and warehouse robots are determined, all tasks are randomly assigned to the available robots, and corresponding task execution sequences are constructed.

Table 1. Schematic representation of chromosome encoding

Task Sequence Index	Picking Station Index	Robot Index
1	1	1
2	4	3
3	5	2
4	2	2
5	3	1
6	2	3
7	4	3
8	5	1
9	3	3
10	2	2

3.2.2 Fitness function

The reciprocal of the total travel distance incurred by warehouse robots during picking operations is adopted as the fitness function, with the objective of identifying the optimal task allocation scheme. The fitness function is defined as follows:

$$f = \frac{1}{\sum_{r=1}^{m} \sum_{b=1}^{n} \sum_{s=1}^{k} x_{rb}^{s} \cdot y_{rb} \cdot d_{rb}^{s}} \tag{7}$$

3.2.3 Improved genetic operations

A roulette wheel selection strategy is employed, in which the probability of an individual being selected is proportional to its fitness value. A two-point crossover operator is adopted. Specifically, two gene positions are randomly selected within a pair of parent chromosomes, and the gene segments between these positions are exchanged to generate offspring chromosomes. After completing genetic operations within the population, simulated annealing-based perturbation is introduced. A gene position is randomly selected, along with the changes of the corresponding manual picking station index and warehouse robot index. The fitness of the perturbed individual is then recalculated. Whether the new solution is accepted is determined according to the Metropolis criterion.

3.3 Task Allocation Procedure Using the Improved Genetic Algorithm

The improved genetic algorithm is used for task allocation of warehouse robots as follows:

Step 1: Target storage racks are identified based on the system's order information. The corresponding task indices and storage rack coordinates are retrieved.

Step 2: Relevant information, including warehouse robots, transportation tasks, and manual picking stations, is encoded using the three-layer matrix encoding. A randomized strategy is used to generate an initial population.

Step 3: Fitness values of individuals are evaluated using the fitness function. The elite preservation incorporated improves the selection operator. Individuals with higher fitness values are retained.

Step 4: The two-point crossover operator applied generates new individuals. Duplicate genes produced after crossover are corrected based on the mapping relationship.

Step 5: Individuals in the population are mutated according to the mutation probability. Any duplicate genes generated during mutation are corrected using the mapping relationship.

Step 6: The simulated annealing introduced perturbs chromosome genes. Whether newly generated solutions are accepted is determined according to the Metropolis criterion.

Step 7: The current iteration number is evaluated against the termination condition. If the condition has not been satisfied, the procedure returns to Step 3 for continued optimization. Otherwise, the algorithm terminates, outputting the optimal solution.

3.4 Simulation Design and Analysis for Task Allocation

In the simulation experiments, the intelligent warehousing environment was modeled using a 30 × 30 grid-based map. After defining the storage rack coordinates, transportation tasks were randomly generated by the system. The system consisted of 5 warehouse robots, 5 manual picking stations, and 196 movable storage racks (2 × 2 grid configuration). A total of 50 transportation tasks were randomly generated. All warehouse robots departed from their initial docking stations and sequentially executed the assigned tasks.

To simplify the analysis, potential positional conflicts among moving robots are not considered, and all robots are assumed to operate at a constant speed. The coordinate information of the transportation tasks is presented in Table 2, and the spatial distribution of task locations is illustrated in Figure 3.

Table 2. Spatial information of transportation tasks

Task No.	Task Coordinates	Storage Rack No.	Task No.	Task Coordinates	Storage Rack No.
b1	(2, 27)	1	b26	(14, 15)	97
b2	(6, 26)	5	b27	(19, 15)	101
b3	(11, 27)	11	b28	(23, 14)	107
b4	(14, 26)	13	b29	(26, 15)	109
b5	(19, 26)	17	b30	(2, 10)	113
b6	(22, 26)	23	b31	(6, 11)	117
b7	(27, 27)	28	b32	(7, 10)	119
b8	(3, 23)	31	b33	(15, 10)	127
b9	(6, 23)	33	b34	(18, 10)	131
b10	(10, 23)	39	b35	(22, 10)	135
b11	(15, 23)	41	b36	(26, 11)	139
b12	(15, 22)	43	b37	(2, 7)	141
b13	(18, 23)	45	b38	(3, 7)	142
b14	(23, 23)	51	b39	(10, 7)	149
b15	(26, 22)	55	b40	(14, 7)	153
b16	(27, 22)	56	b41	(18, 7)	157
b17	(2, 18)	57	b42	(19, 6)	159
b18	(7, 19)	63	b43	(27, 7)	167
b19	(11, 18)	65	b44	(2, 2)	169
b20	(14, 19)	71	b45	(6, 3)	173
b21	(27, 19)	83	b46	(11, 3)	177
b22	(26, 18)	81	b47	(10, 2)	179
b23	(6, 15)	89	b48	(19, 3)	185
b24	(6, 14)	91	b49	(23, 2)	191
b25	(10, 14)	95	b50	(26, 3)	195

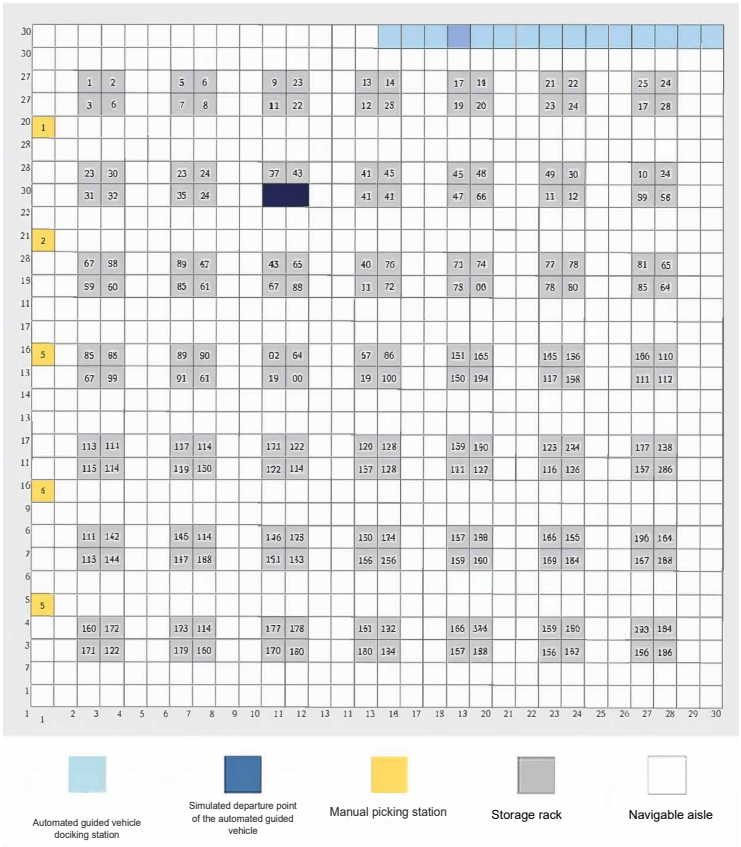


Figure 3. Schematic representation of transportation task locations

In the simulation experiments, the population size of the genetic algorithm was set to 100, the crossover probability P_c was set to 0.8, the mutation probability P_m was set to 0.08, and the maximum number of iterations was set to 500. A total of 50 tasks and 5 warehouse robots were considered.

The task allocation problem was solved using both the conventional genetic algorithm and the improved genetic algorithm for comparative analysis. Figure 4 and Figure 5 show the convergence curves of the conventional and improved genetic algorithms. Figure 6 and Figure 7 show the robot travel paths under the two algorithms.

Table 3 shows the comparative results between the two genetic algorithms.

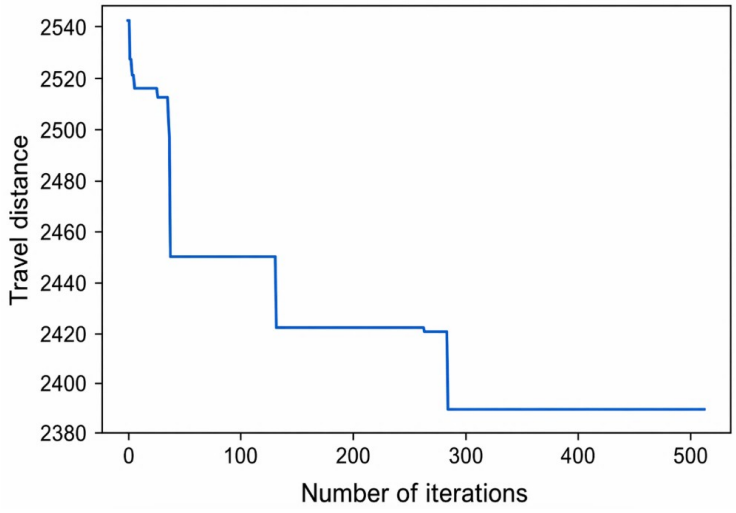


Figure 4. Convergence curve using the conventional genetic algorithm

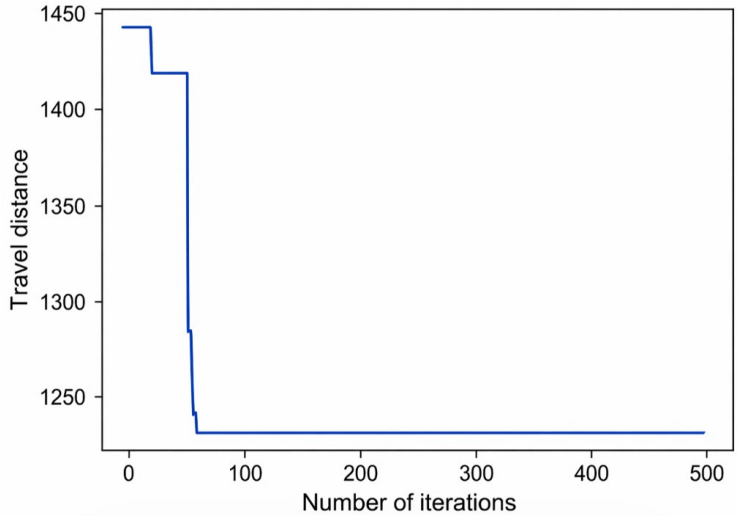


Figure 5. Convergence curve using the improved genetic algorithm

Table 3. Performance comparison between the conventional and improved genetic algorithms

Algorithm	Convergence Generation	Travel Distance (m)
Conventional genetic algorithm	286	2385.08
Improved genetic algorithm	70	1227.63

According to the simulation results, the improved genetic algorithm achieves convergence after 70 generations, compared to the conventional genetic algorithm (286 generations). This demonstrates a substantial enhancement in algorithmic efficiency, a 75.52% increase in convergence speed. The improved genetic algorithm obtains a total travel distance of 1227.63 m, compared with 2385.08 m for the conventional genetic algorithm. This is a reduction of 48.53%. This significant decrease improves operational efficiency in picking tasks.

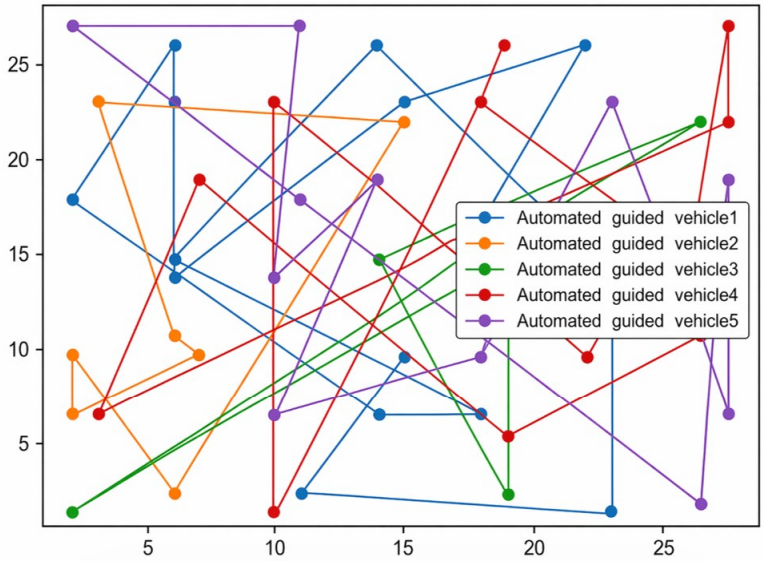


Figure 6. Robot travel paths using the conventional genetic algorithm

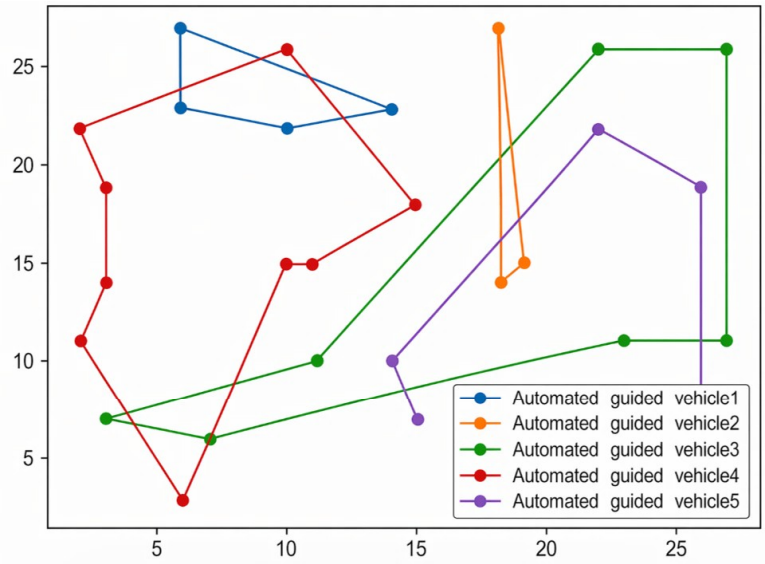


Figure 7. Robot travel paths using the improved genetic algorithm

Furthermore, the improved algorithm is capable of achieving a more balanced and reasonable distribution of operational tasks among multiple warehouse robots, effectively avoiding unnecessary path detours and redundant movements during robot execution. This fully demonstrates that the overall task execution sequence generated by the algorithm is well-structured, logically organized, and highly conducive to efficient and orderly operation of the entire warehouse system.

In addition, the task allocation strategy derived from the proposed improved method exhibits superior effectiveness and rationality compared with the conventional genetic algorithm under the same experimental conditions. On the whole, the improved genetic algorithm not only converges more rapidly during the iterative optimization process but also greatly reduces the total travel distance and path cost of warehouse robots. Consequently, these advantages jointly contribute to more balanced and reasonable task allocation among multiple robots, as well as significantly enhanced operational efficiency, execution stability and overall working performance of the robot system.

4 Path Planning Optimization Model and Algorithm

Path planning for warehouse robots is critical in improving the operational efficiency of intelligent warehousing systems. An improved A* algorithm is employed to generate optimal paths for warehouse robots between multiple starting and target locations. The planned paths should avoid all obstacles and enable robots to execute picking tasks safely and reach their designated destinations.

4.1 Improved A* Algorithm

The A* algorithm plans paths through a search mechanism by progressively exploring neighboring nodes of the initial node. An evaluation function is employed to assess the quality of candidate path nodes. In this way, the optimal path within the map environment can be identified [14]. The evaluation function is critical in the A* algorithm because it determines each node's path cost and evaluates path quality [15].

In this study, the Manhattan distance is adopted as the heuristic function for estimating the path cost. The Manhattan distance is the sum of the absolute differences between the horizontal and vertical coordinates of two points in a coordinate system. It is used to estimate the distance between the current node n and the target node M. The heuristic cost function is expressed as follows:

$$\mathrm{h}(n) = |x_n - x_m| + |y_n - y_m| \tag{8}$$

where, (x_n, y_n) and (x_m, y_m) denote the coordinates of the current node n and the target node M.

To reduce the number of turning maneuvers of robots, a turning cost is introduced into the evaluation function of the A* algorithm. The improved evaluation function is defined as follows:

$$f(n) = g(n) + \mathrm{h}(n) + C_{\text{turn}} \cdot m \tag{9}$$

where, C_{turn} represents the turning cost coefficient, and m denotes the number of turning maneuvers. Based on experimental analysis, $C_{\text{turn}} = 0.6$ can minimize the number of turning maneuvers.

4.2 Path Planning Procedure Using the Improved A* Algorithm

The path planning problem for warehouse robots is addressed using the improved A* algorithm through the following procedure:

Step 1: Based on the task allocation results obtained in Section 3, a complete picking task chain is constructed. The sequence and spatial coordinates of each node within the task chain are determined, such that they can be sequentially treated as the start nodes and target nodes in the path planning process.

Step 2: The warehousing environment is simplified into a two-dimensional grid-based map. The coordinates of the robot's start node and target node are specified. Two lists—OpenList and CloseList—are initialized to store nodes to be explored and nodes that have already been evaluated, respectively.

Step 3: The start node S is inserted into the OpenList, and the CloseList is initialized as empty. A search is then initiated from the start node by expanding its four-connected neighboring nodes. Valid nodes—defined as non-obstacle nodes that are not contained in the CloseList—are identified and processed, and the start node S is subsequently moved to the CloseList.

Step 4: If the OpenList is examined to be empty, the path search is considered unsuccessful. Otherwise, the procedure proceeds to the next step.

Step 5: All OpenList nodes are traversed and evaluated for validity. Those that are identified as invalid are excluded from further expansion. Then the modified path cost $f(n)$ is computed for valid nodes. The node i with the minimum $f(n)$ value is selected as the next parent node. If node i corresponds to the target node, the path search is completed and the optimal path is obtained. Otherwise, the procedure continues to the next step.

Step 6: Node i is transferred to the CloseList, and its four-connected neighboring nodes are expanded as child nodes. The validity of those child nodes is evaluated. Invalid nodes are excluded from further expansion. If a valid child node is already present in the OpenList, the procedure proceeds to Step 7; otherwise, it proceeds to Step 8.

Step 7: The actual path cost $g(n)$ concerning reaching the node via the current path is compared with the existing value. If a lower value is obtained, the parent node of the corresponding node is updated to node i. The procedure then proceeds to Step 9.

Step 8: The child node is inserted into the OpenList, its parent node is recorded, and its modified path cost $f(n)$ is computed. The procedure then proceeds to Step 9.

Step 9: If the target node has been identified, the search process is terminated and the path planning result is output. Otherwise, the procedure returns to Step 4 to continue the search for the optimal path.

4.3 Simulation Design and Analysis for Path Planning

To evaluate the effectiveness of the improved A* algorithm in path planning, four groups of simulation experiments were conducted. Both the conventional and improved A* algorithms were applied, and their performance was comparatively analyzed.

In the first simulation scenario, the grid indices of the start and target nodes are 239 and 721, with coordinates (28, 22) and (0, 5). In the second simulation scenario, the start node corresponds to grid index 836 with coordinates (25, 2), and the target node corresponds to grid index 272 with coordinates (1, 20). In the third simulation scenario, the start node corresponds to grid index 715 with coordinates (24, 6). In the fourth simulation scenario, the start

node corresponds to grid index 634 with coordinates (8, 3), and the target node corresponds to grid index 19 with coordinates (18, 29). The turning cost coefficient C_{turn} was set to 0.6. Path planning was then performed using both the conventional A* algorithm and the improved A* algorithm for warehouse robots. The resulting path planning performance is summarized in Table 4.

Table 4. Path planning results

Experimental Group	Path Length	Turning Maneuvers (Conventional A*)	Turning Maneuvers (Improved A*)
Group 1	45	17	3
Group 2	42	17	4
Group 3	29	11	3
Group 4	36	16	3

The schematic representations of the path planning results for the four simulation scenarios are illustrated in Figure 8.

The results of the four experimental scenarios indicate that the improved A* algorithm significantly reduces the number of turning maneuvers without increasing the path length. Specifically, the average number of turning maneuvers decreases from 15.25 to 3.25 (a reduction of 78.7%). The decrease in directional changes is substantial. The improved A* algorithm generates feasible paths from the start node to the target node while maintaining the optimal path length. At the same time, the minimized number of turning maneuvers leads to smoother paths. Consequently, time loss and energy consumption due to frequent turning are significantly reduced. This enhances the operational efficiency of warehouse robots in picking tasks.



Figure 8. Schematic representations of path planning results across four experimental groups

5 Conclusion

This study optimized the operational efficiency of intelligent warehousing systems for e-commerce fulfillment centers. Task allocation and path planning were jointly optimized using an improved genetic algorithm integrated with an enhanced A* algorithm. This effectively addressed key challenges in multi-robot scheduling, including redundant travel distance and excessive turning maneuvers. The proposed method provides a theoretical and practical foundation for improving the operational efficiency of intelligent warehousing systems. In the future, real-world industrial requirements can be further incorporated to facilitate the deployment of the proposed algorithms in practical logistics systems, which is expected to further support the sustainable development of smart logistics.

Author Contributions

Conceptualization, L.L.; methodology, L.L. and Q.L.; software, Q.L. and Z.H.; validation, Q.L. and Z.H.; formal analysis, Q.L. and Z.H.; investigation, Z.H.; resources, L.L.; data curation, Q.L. and Z.H.; writing—original draft preparation, Z.H.; writing—review and editing, L.L. and Q.L.; visualization, Q.L.; supervision, L.L.; project administration, L.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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